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Capacitive ultrasonic transducer device, manufacturing method thereof and transducer array

Abstract

A capacitive ultrasonic transducer device includes a substrate, a first capacitive structure, a second capacitive structure, a first film structure and a second film structure. The first capacitive structure is disposed on the substrate, and includes a first electrode and a second electrode. A first gap and a dielectric layer are located between the first electrode and the second electrode. The second capacitive structure is disposed on the substrate, and includes a third electrode and a fourth electrode. A second gap is located between the third electrode and the fourth electrode. The first film structure is configured to seal the first gap. The second film structure is connected to the third electrode and the fourth electrode, and configured to seal the second gap. A first width between the first electrode and the second electrode is different from a second width of the second gap.

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Background/Summary

RELATED APPLICATIONS

(1) This application claims priority to Taiwan Application Serial Number 111136868, filed Sep. 28, 2022, which is herein incorporated by reference.

BACKGROUND

Technical Field

(2) The present disclosure relates to a transducer device, a manufacturing method thereof and a transducer array. More particularly, the present disclosure relates to a capacitive ultrasonic transducer device, a manufacturing method thereof and a transducer array.

Description of Related Art

(3) The conventional capacitive micromachined ultrasonic transducer (CMUT) integrates the micro-electromechanical devices of the CMUT with the circuit by wafer bonding or packaging. Due to a driving circuit of the capacitive transducer need to be driven by a high voltage, the development cost of the CMUT developed by System on a Chip (SoC) is higher, thereby increasing the cost of the micro-electromechanical system (MEMS) with large area.

(4) Thus, a capacitive ultrasonic transducer device, which manufactures the micro-electromechanical capacitive ultrasonic elements on a Complementary Metal Oxide Semiconductor (CMOS) by a lost cost process, is commercially desirable.

SUMMARY

(5) According to one aspect of the present disclosure, a capacitive ultrasonic transducer device includes a substrate, a first capacitive structure, a second capacitive structure, a first film structure and a second film structure. The first capacitive structure is disposed on the substrate, and includes a first electrode and a second electrode. A first gap and a dielectric layer are located between the first electrode and the second electrode. The second capacitive structure is disposed on the substrate, and includes a third electrode and a fourth electrode. A second gap is located between the third electrode and the fourth electrode. The first film structure is connected to the first electrode and the dielectric layer, and configured to seal the first gap. The second film structure is connected to the third electrode and the fourth electrode, and configured to seal the second gap. A first width between the first electrode and the second electrode is different from a second width of the second gap.

(6) According to another aspect of the present disclosure, a transducer array includes a plurality of capacitive ultrasonic transducer devices and an interface circuit. At least one of the capacitive ultrasonic transducer devices includes a substrate, a first capacitive structure, a second capacitive structure, a first film structure and a second film structure. The first capacitive structure is disposed on the substrate, and includes a first electrode and a second electrode. A first gap and a dielectric layer are located between the first electrode and the second electrode. The second capacitive structure is disposed on the substrate, and includes a third electrode and a fourth electrode. A second gap is located between the third electrode and the fourth electrode. The first film structure is connected to the first electrode and the dielectric layer, and configured to seal the first gap. The second film structure is connected to the third electrode and the fourth electrode, and configured to seal the second gap. The interface circuit is for the capacitive ultrasonic transducer devices stacking thereon, and electrically connected to the capacitive ultrasonic transducer devices. A first width between the first electrode and the second electrode is different from a second width of the second gap.

(7) According to further another aspect of the present disclosure, a manufacturing method of a

capacitive ultrasonic transducer device includes performing a printing step, a first etching step, a second etching step, a third etching step and a film disposing step. The printing step is performed to print a first capacitive precursor structure and a second capacitive precursor structure on a substrate. The first etching step is performed to etch a metallic compound of the first capacitive precursor structure and a metallic compound of the second capacitive precursor structure to form a plurality of openings according to an isotropic wet etching process. The second etching step is performed to etch a dielectric layer of the first capacitive precursor structure and a dielectric layer of the second capacitive precursor structure from the openings according to an anisotropy dry etching process. The third etching step is performed to etch an aluminum copper alloy of the first capacitive precursor structure and an aluminum copper alloy of the second capacitive precursor structure to form a first capacitive structure and a second capacitive structure, respectively, according to the isotropic wet etching process. The film disposing step is performed to dispose a first film structure and a second film structure on a surface of the first capacitive structure and a surface of the second capacitive structure, respectively.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows:
- (2) FIG. 1 shows a schematic view of a capacitive ultrasonic transducer device according to a first embodiment of the present disclosure.
- (3) FIG. 2 shows a cross-sectional view of the capacitive ultrasonic transducer device along line 2-2 of FIG. 1.
- (4) FIG. 3 shows a cross-sectional view of a capacitive ultrasonic transducer device according to a second embodiment of the present disclosure.
- (5) FIG. 4 shows a schematic view of a transducer array according to a third embodiment of the present disclosure.
- (6) FIG. 5 shows a flow chart of a manufacturing method of a capacitive ultrasonic transducer device according to a fourth embodiment of the present disclosure.
- (7) FIG. 6 shows a schematic view of a printing step of the manufacturing method of the capacitive ultrasonic transducer device according to the embodiment in FIG. 5.
- (8) FIG. 7 shows a schematic view of a first etching step of the manufacturing method of the capacitive ultrasonic transducer device according to the embodiment in FIG. 5.
- (9) FIG. 8 shows a schematic view of a second etching step of the manufacturing method of the capacitive ultrasonic transducer device according to the embodiment in FIG. 5.
- (10) FIG. 9 shows a schematic view of a third etching step of the manufacturing method of the capacitive ultrasonic transducer device according to the embodiment in FIG. 5.
- (11) FIG. 10 shows a schematic view of a film disposing step of the manufacturing method of the capacitive ultrasonic transducer device according to the embodiment in FIG. 5.
- (12) FIG. 11 shows a flow chart of a manufacturing method of a capacitive ultrasonic transducer device according to a fifth embodiment of the present disclosure.
- (13) FIG. 12 shows a schematic view of a laser step of the manufacturing method of the capacitive ultrasonic transducer device according to the embodiment in FIG. 11.
- (14) FIG. 13 shows a schematic view of a wiring step of the manufacturing method of the capacitive ultrasonic transducer device according to the embodiment in FIG. 11.

DETAILED DESCRIPTION

- (15) The embodiment will be described with the drawings. For clarity, some practical details will be described below. However, it should be noted that the present disclosure should not be limited

by the practical details, that is, in some embodiment, the practical details is unnecessary. In addition, for simplifying the drawings, some conventional structures and elements will be simply illustrated, and repeated elements may be represented by the same labels.

(16) It will be understood that when an element (or device) is referred to as be “connected to” another element, it can be directly connected to other element, or it can be indirectly connected to the other element, that is, intervening elements may be present. In contrast, when an element is referred to as be “directly connected to” another element, there are no intervening elements present. In addition, the terms first, second, third, etc. are used herein to describe various elements or components, these elements or components should not be limited by these terms. Consequently, a first element or component discussed below could be termed a second element or component.

(17) Please refer to FIG. 1 and FIG. 2. FIG. 1 shows a schematic view of a capacitive ultrasonic transducer device **100** according to a first embodiment of the present disclosure. FIG. 2 shows a cross-sectional view of the capacitive ultrasonic transducer device **100** along line 2-2 of FIG. 1. The capacitive ultrasonic transducer device **100** includes a substrate **110**, a first capacitive structure **120**, a second capacitive structure **130**, a first film structure **140** and a second film structure **150**. The first capacitive structure **120** is disposed on the substrate **110**, and includes a first electrode **121** and a second electrode **122**. A first gap **123** and a dielectric layer **124** are located between the first electrode **121** and the second electrode **122**. The second capacitive structure **130** is disposed on the substrate **110**, and includes a third electrode **131** and a fourth electrode **132**. A second gap **133** is located between the third electrode **131** and the fourth electrode **132**. The first film structure **140** is connected to the first electrode **121** and the dielectric layer **124**, and configured to seal the first gap **123**. The second film structure **150** is connected to the third electrode **131** and the fourth electrode **132**, and configured to seal the second gap **133**. A first width G1 between the first electrode **121** and the second electrode **122** is different from a second width G2 of the second gap **133**. Thus, the capacitive ultrasonic transducer device **100** of the present disclosure can provide different capacitive transduction gaps with different widths (i.e., the first width G1 and the second width G2) so as to increase the application flexibility of a capacitive ultrasonic element.

(18) Moreover, the substrate **110** can be a metal connecting layer of a Complementary Metal Oxide Semiconductor (CMOS). The first capacitive structure **120** can be a Metal-Inductor-Metal (MIM) structure. The second capacitive structure **130** can be a Titanium Nitride (TiN) capacitive structure. Each of the first electrode **121**, the third electrode **131** and the fourth electrode **132** can be TiN. The second electrode **122** can be a metal layer of a sandwich structure. The dielectric layer **124** can be located on the second electrode **122**, and a thickness of the dielectric layer **124** can be 40 nanometer (nm). Each of the first film structure **140** and the second film structure **150** can be a dielectric material or film formed by a thin film deposition process. The first film structure **140** is connected from each of the two ends of the first electrode **121** to the dielectric layer **124**, thereby sealing the first gap **123**. The second film structure **150** is connected from each of the two ends of the third electrode **131** to the fourth electrode **132**, thereby sealing the second gap **133**. Thus, the capacitive ultrasonic transducer device **100** of the present disclosure has different capacitive structures (i.e., the MIM structure and the TiN capacitive structure) at the same time.

(19) In the first embodiment, the substrate **110** can be a metal connecting layer on a 0.18 micrometer (μm) CMOS platform, the first capacitive structure **120** can be a Capacitor-Top-Metal (CTM) between a M5 metal layer and a M6 metal layer, the first electrode **121** can be TiN of the aforementioned CTM, the second electrode **122** can be the M5 metal layer, the second capacitive structure **130** can be a M4 metal layer, the third electrode **131** and the fourth electrode **132** can be two TiN layers of the M4 metal layer, but the present disclosure is not limited thereto. Moreover, the capacitive ultrasonic transducer device **100** of the present disclosure utilizes the structure over the M4 metal layer to form the first capacitive structure **120** and the second capacitive structure **130**, and reserves the structure under the M4 metal layer to increase the programming flexibility of the circuit.

(20) In detail, by sealing the first capacitive structure **120** and the second capacitive structure **130** via the first film structure **140** and the second film structure **150**, respectively, the capacitive ultrasonic transducer device **100** can be applied to one of a solid-state environment and a liquid environment. In the first embodiment, the capacitive ultrasonic transducer device **100** can be a fingerprint sensing device, a mass sensing device disposed under water or disposed in the electrolytic solution, but the present disclosure is not limited thereto. Moreover, by sealing the first gap **123** and the second gap **133** via the first film structure **140** and the second film structure **150**, respectively, the first gap **123** and the second gap **133** are under a vacuum state. In the vacuum state, the pressure is relatively low, the air damping can be decreased, and the Q factor can be increased. Therefore, the capacitive ultrasonic transducer device **100** of the present disclosure can have sufficient electromechanical conversion performance while disposing in normal atmosphere environment and the liquid environment.

(21) Furthermore, the first capacitive structure **120** can be a receiver, and the second capacitive structure **130** can be a transmitter. The first width G1 is less than the second width G2. The first width G1 is greater than or equal to 100 nm, and less than or equal to 150 nm. The second width G2 is greater than or equal to 350 nm, and less than or equal to 450 nm. In the first embodiment, the first width G1 can be 120 nm, the second width G2 can be 450 nm, but the present disclosure is not limited thereto. Because the first width G1 of the first capacitive structure **120** is narrow, the first capacitive structure **120** can provide great electromechanical conversion efficiency, and the first capacitive structure **120** can also have great electromechanical transduction efficiency without driven by a high voltage. Moreover, the receiver (i.e., the first capacitive structure **120**) has high sensitivity. The second width G2 of the second capacitive structure **130** is wider, and the film displacement generated by the second capacitive structure **130** is greater. Thus, the capacitive ultrasonic transducer device **100** of the present disclosure can have high detecting sensitivity and high displacement, and also can be driven by low voltage (under 25 V).

(22) Please refer to FIG. 1, a thickness h of the first film structure **140** and a thickness h of the second film structure **150** of the capacitive ultrasonic transducer device **100** can be both greater than or equal to 0.5 μm , and less than or equal to 5 μm . The etching width W e can be greater than or equal to 5 μm , and less than or equal to 20 μm . In the first embodiment, a length L of the capacitive ultrasonic transducer device **100** can be greater than or equal to 30 μm , and less than or equal to 100 μm . The capacitive ultrasonic transducer device **100** can etch from an edge, which has a lower oscillation speed.

(23) Please refer to FIG. 1 to FIG. 3. FIG. 3 shows a cross-sectional view of a capacitive ultrasonic transducer device **100a** according to a second embodiment of the present disclosure. The capacitive ultrasonic transducer device **100a** includes a substrate **110**, a first capacitive structure **120**, a second capacitive structure **130**, a first film structure **140** and a second film structure **150**. In the second embodiment, the substrate **110**, a first capacitive structure **120**, the second capacitive structure **130**, the first film structure **140** and the second film structure **150** of the capacitive ultrasonic transducer device **100a** are the same as the substrate **110**, a first capacitive structure **120**, the second capacitive structure **130**, the first film structure **140** and the second film structure **150** of the capacitive ultrasonic transducer device **100**, respectively, and will not be described again herein. The capacitive ultrasonic transducer device **100a** can further include two conductive structures **170** and two isolation layers **180**. The two conductive structures **170** are electrically connected to the first capacitive structure **120** and the second capacitive structure **130**, respectively. The two isolation layers **180** cover the two conductive structures **170**, respectively. Thus, the capacitive ultrasonic transducer device **100a** can be disposed under water or in the electrolytic solution.

(24) Please refer to FIG. 2, FIG. 3 and FIG. 4. FIG. 4 shows a schematic view of a transducer array **300** according to a third embodiment of the present disclosure. The transducer array **300** includes a plurality of capacitive ultrasonic transducer devices **100** and an interface circuit **200**. In the third embodiment, each of the capacitive ultrasonic transducer devices **100** can be the same as the

capacitive ultrasonic transducer device **100** in the first embodiment or the capacitive ultrasonic transducer device **100a** in the second embodiment, and will not be described again herein. The interface circuit **200** is for the capacitive ultrasonic transducer devices **100** stacking thereon, and electrically connected to the capacitive ultrasonic transducer devices **100**. Thus, the area and the wiring complexity of the transducer array **300** can be reduced by manufacturing the capacitive ultrasonic transducer devices **100** on the CMOS, and disposing the interface circuit **200** under the capacitive ultrasonic transducer devices **100** vertically.

(25) Moreover, due to the boundary condition of the peripheral section **310** of the transducer array **300** is different from the capacitive ultrasonic transducer devices **100**, the natural frequency of the peripheral section **310** is different from the natural frequency of the capacitive ultrasonic transducer devices **100**. Therefore, the capacitive ultrasonic transducer devices **100** are not disposed on the peripheral section **310** of the transducer array **300**.

(26) Please refer to FIG. 2 and FIG. 5 to FIG. 10. FIG. 5 shows a flow chart of a manufacturing method **S10** of a capacitive ultrasonic transducer device **100** according to a fourth embodiment of the present disclosure. FIG. 6 shows a schematic view of a printing step **S01** of the manufacturing method **S10** of the capacitive ultrasonic transducer device **100** according to the embodiment in FIG. 5. FIG. 7 shows a schematic view of a first etching step **S02** of the manufacturing method **S10** of the capacitive ultrasonic transducer device **100** according to the embodiment in FIG. 5. FIG. 8 shows a schematic view of a second etching step **S03** of the manufacturing method **S10** of the capacitive ultrasonic transducer device **100** according to the embodiment in FIG. 5. FIG. 9 shows a schematic view of a third etching step **S04** of the manufacturing method **S10** of the capacitive ultrasonic transducer device **100** according to the embodiment in FIG. 5. FIG. 10 shows a schematic view of a film disposing step **S05** of the manufacturing method **S10** of the capacitive ultrasonic transducer device according to the embodiment in FIG. 5. The manufacturing method **S10** of the capacitive ultrasonic transducer device **100** includes performing the printing step **S01**, the first etching step **S02**, the second etching step **S03**, the third etching step **S04** and the film disposing step **S05**.

(27) In the printing step **S01**, a first capacitive precursor structure **20** and a second capacitive precursor structure **30** are printed on a substrate **110**.

(28) In the first etching step **S02**, the metallic compounds **21**, **22** of the first capacitive precursor structure **20** and the metallic compounds **31**, **32** of the second capacitive precursor structure **30** are etched to form a plurality of openings **11**, **12**, **13**, **14** according to an isotropic wet etching process. In detail, the isotropic wet etching process etches the metallic compounds **21**, **22**, **31**, **32** via an etching solution with H.sub.2SO.sub.4 and H.sub.2O.sub.2, but the present disclosure is not limited thereto.

(29) In the second etching step **S03**, the dielectric layers **23**, **24** of the first capacitive precursor structure **20** and the dielectric layers **33**, **34** of the second capacitive precursor structure **30** are etched from the openings **11**, **12**, **13**, **14** according to an anisotropy dry etching process. In detail, the anisotropy dry etching process can be a Reactive Ion Etching (RIE), but the present disclosure is not limited thereto.

(30) In the third etching step **S04**, an aluminum copper alloy 25 of the first capacitive precursor structure **20** and an aluminum copper alloy 35 of the second capacitive precursor structure **30** are etched to form a first capacitive structure **120** (as shown in FIG. 9) and a second capacitive structure **130** (as shown in FIG. 9), respectively, according to the isotropic wet etching process.

(31) In the film disposing step **S05**, a first film structure **140** and a second film structure **150** are disposed on a surface of the first capacitive structure **120** and a surface of the second capacitive structure **130**, respectively to form the capacitive ultrasonic transducer device **100** in FIG. 2. In the fourth embodiment, the capacitive ultrasonic transducer device **100** can be the same as the capacitive ultrasonic transducer device **100** in the first embodiment, and the present disclosure is not limited thereto.

(32) Thus, the manufacturing method **S10** of the capacitive ultrasonic transducer device **100** can manufacture the first capacitive structure **120** and the second capacitive structure **130** of the capacitive ultrasonic transducer device **100** on the CMOS at the same time by same manufacturing process (i.e., the isotropic wet etching process and the anisotropy dry etching process) with low cost, thereby decreasing the manufacturing cost of the capacitive ultrasonic system.

(33) Please refer to FIG. 1, FIG. 3, FIG. 5 and FIG. 11 to FIG. 13. FIG. 11 shows a flow chart of a manufacturing method **S10a** of a capacitive ultrasonic transducer device **100a** according to a fifth embodiment of the present disclosure. FIG. 12 shows a schematic view of a laser step **S06** of the manufacturing method **S10a** of the capacitive ultrasonic transducer device **100a** according to the embodiment in FIG. 11. FIG. 13 shows a schematic view of a wiring step **S07** of the manufacturing method **S10a** of the capacitive ultrasonic transducer device **100a** according to the embodiment in FIG. 11. The manufacturing method **S10a** of the capacitive ultrasonic transducer device **100a** includes performing a printing step **S01**, a first etching step **S02**, a second etching step **S03a**, a third etching step **S04**, a film disposing step **S05**, the laser step **S06**, the wiring step **S07** and an isolation layer disposing step **S08**. In the fifth embodiment, the printing step **S01**, the first etching step **S02**, the third etching step **S04** and the film disposing step **S05** are the same as the printing step **S01**, the first etching step **S02**, the third etching step **S04** and the film disposing step **S05** in the fourth embodiment, respectively, and will not be described again herein. The manufacturing method **S10a** of the capacitive ultrasonic transducer device **100a** can further include the laser step **S06**, the wiring step **S07** and the isolation layer disposing step **S08**. The second etching step **S03a** can include a frequency matching step **S031**.

(34) In the frequency matching step **S031**, a resonance frequency of the first capacitive structure **120** and a resonance frequency of the second capacitive structure **130** are adjusted according to an etching time of the second etching step **S03a**. Moreover, the frequency matching step **S031** can adjust a thickness of the dielectric layer **27** (as shown in FIG. 8 and FIG. 9) to match a natural frequency of the first capacitive structure **120** and a natural frequency of the second capacitive structure **130** by extending or shortening the etching time of the second etching step **S03a**.

(35) In other embodiment, the manufacturing method of the capacitive ultrasonic transducer device can adjust a bias applied on one of the first capacitive structure and the second capacitive structure to match a natural frequency of the first capacitive structure and a natural frequency of the second capacitive structure according to an electrical soft spring effect. Because the first width of the first capacitive structure is narrower than the second width of the second capacitive structure, the first capacitive structure provides wider frequency adjusting range.

(36) In the laser step **S06**, a partial dielectric layer **26** (as shown in FIG. 10) of the first capacitive structure **120** and a partial dielectric layer **36** (as shown in FIG. 10) of the second capacitive structure **130** are removed to form two conductive layers **160** according to a laser process.

(37) In the wiring step **S07**, two conductive structures **170** are disposed on the two conductive layers **160** of the first capacitive structure **120** and the second capacitive structure **130**, respectively. The two conductive structures **170** are electrically connected to the first capacitive structure **120** and the second capacitive structure **130**, respectively.

(38) In the isolation layer disposing step **S08**, the two conductive structures **170** are covered by two isolation layers **180**, respectively, and the capacitive ultrasonic transducer device **100a** (as shown in FIG. 3) is formed. In the fifth embodiment, the capacitive ultrasonic transducer device **100a** can be the same as the capacitive ultrasonic transducer device **100a** in the second embodiment, but the present disclosure is not limited thereto.

(39) According to the aforementioned embodiments and examples, the advantages of the present disclosure are described as follows. 1. The capacitive ultrasonic transducer device of the present disclosure can provide different capacitive transduction gaps with different widths (i.e., the first width and the second width) so as to increase the application flexibility of a capacitive ultrasonic element. 2. The area and the wiring complexity of the transducer array can be reduced by

manufacturing the capacitive ultrasonic transducer devices **100** on the CMOS, and disposing the interface circuit under the capacitive ultrasonic transducer devices vertically. 3. The manufacturing method of the capacitive ultrasonic transducer device can manufacture the first capacitive structure and the second capacitive structure of the capacitive ultrasonic transducer device on the CMOS at the same time by same manufacturing process (i.e., the isotropic wet etching process and the anisotropy dry etching process) with low cost, thereby decreasing the manufacturing cost of the capacitive ultrasonic system.

(40) Although the present disclosure has been described in considerable detail with reference to certain embodiments thereof, other embodiments are possible. Therefore, the spirit and scope of the appended claims should not be limited to the description of the embodiments contained herein.

(41) It will be apparent to those skilled in the art that various modifications and variations can be made to the structure of the present disclosure without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the present disclosure cover modifications and variations of this disclosure provided they fall within the scope of the following claims.

Claims

1. A capacitive ultrasonic transducer device, comprising: a substrate; a first capacitive structure disposed on the substrate, and comprising: a first electrode; and a second electrode, a first gap and a dielectric layer located between the first electrode and the second electrode; a second capacitive structure disposed on the substrate, and comprising: a third electrode; and a fourth electrode, a second gap located between the third electrode and the fourth electrode; a first film structure connected to the first electrode and the dielectric layer, and configured to seal the first gap; and a second film structure connected to the third electrode and the fourth electrode, and configured to seal the second gap; wherein a first width between the first electrode and the second electrode is different from a second width of the second gap.
2. The capacitive ultrasonic transducer device of claim 1, wherein the first capacitive structure is a receiver, the second capacitive structure is a transmitter, and the first width is less than the second width.
3. The capacitive ultrasonic transducer device of claim 1, wherein the first width is greater than or equal to 100 nanometer (nm), and less than or equal to 150 nm, and the second width is greater than or equal to 350 nm, and less than or equal to 450 nm.
4. The capacitive ultrasonic transducer device of claim 1, wherein the first capacitive structure is a metal-inductor-metal (MIM) structure, and the second capacitive structure is a titanium nitride (TIN) capacitive structure.
5. The capacitive ultrasonic transducer device of claim 1, wherein the capacitive ultrasonic transducer device is applied to one of a solid-state environment and a liquid environment.
6. The capacitive ultrasonic transducer device of claim 1, wherein a thickness of the first film structure and a thickness of the second film structure are both greater than or equal to 0.5 micrometer (μm) and less than or equal to 5 μm .
7. The capacitive ultrasonic transducer device of claim 1, further comprising: two conductive structures electrically connected to the first capacitive structure and the second capacitive structure, respectively; and two isolation layers covering the two conductive structures, respectively.
8. A transducer array, comprising: a plurality of capacitive ultrasonic transducer devices, at least one of the capacitive ultrasonic transducer devices comprising: a substrate; a first capacitive structure disposed on the substrate, and comprising: a first electrode; and a second electrode, a first gap and a dielectric layer located between the first electrode and the second electrode; a second capacitive structure disposed on the substrate, and comprising: a third electrode; and a fourth electrode, a second gap located between the third electrode and the fourth electrode; a first film structure connected to the first electrode and the dielectric layer, and configured to seal the first

gap; and a second film structure connected to the third electrode and the fourth electrode, and configured to seal the second gap; and an interface circuit for the capacitive ultrasonic transducer devices stacking thereon, and electrically connected to the capacitive ultrasonic transducer devices; wherein a first width between the first electrode and the second electrode is different from a second width of the second gap.
